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micrometer = micron = μm

Casting:

→ uses soft, elastic polymers, commonly polydimethylsiloxane (PDMS)

→ A rigid master template (often from silicon) is first patterned using traditional photolithograph

- silicon wafer removed of dust, oil
- liquid light-sensitive chemical (photoresist) is spun on surface into thin layer
- photomask (physical stencil) placed/aligned over coated silicon wafer
- UV light shines through clear part of the photomask onto the photoresist
- chemical wash dissolves either the exposed or unexposed parts of the photoresist (leaving physical pattern)

→ Now, we have our master mold, liquid PDMS is poured over master, cured with heat, and peeled off to form a replica

→ PDMS layers permanently sealed with glass or other polymer substrate to create closed system.

